Swifts and the selection of urban habitats: a citizen science project to increase the ecological knowledge of the species in the Bergamo and Varese provinces

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Anthropization and urbanization

INTRODUCTION

The importance of knowing the conservation status of the Swifts in our territory prompted us to design the project

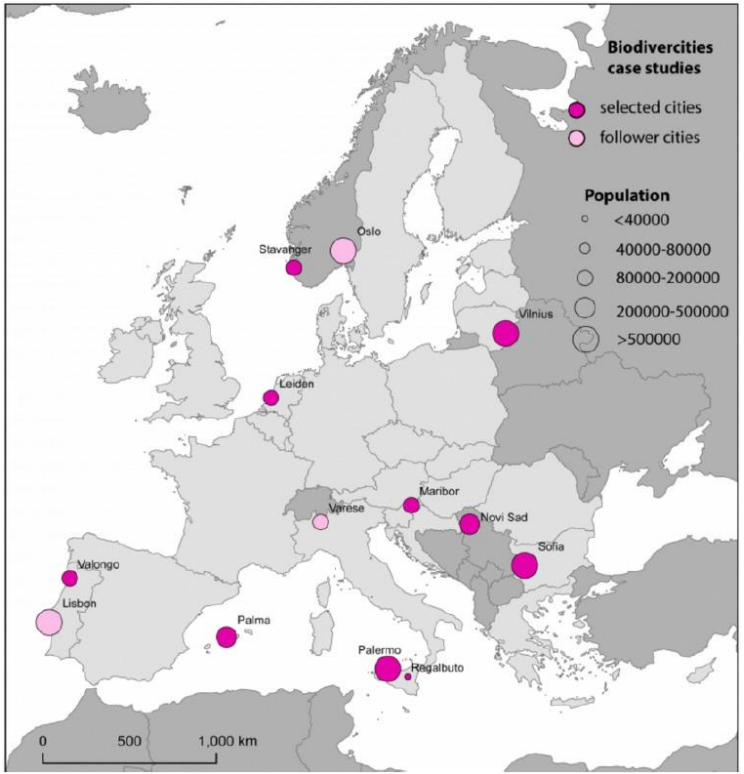
Sulle ali dei rondoni (On the wings of Swifts)

This started with the collaboration of two associations and continued with other organizations and European Commission

INTRODUCTION

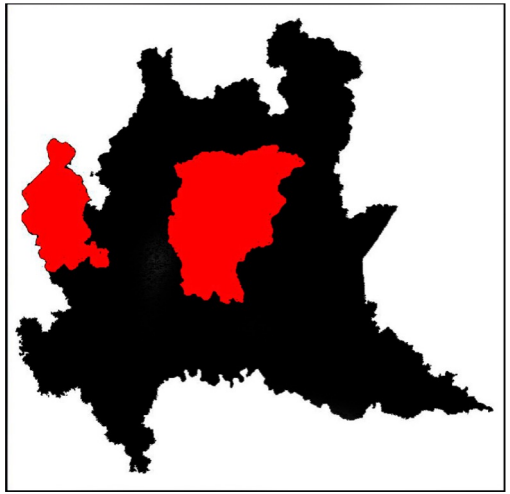
Our project was selected for the **BiodiverCities** program:

We were involved in the European program together with other important cities



OBJECTIVES

- Map of the colonies
- Improving scientific knowledge (eg. Which are the most selected nest sites? Which are the most important buildings?)
- Supporting conservation initiatives
- Raising people's awareness



METHODS

- Citizen science: 100 volunteers, schools, students, municipalities
- Training meetings to identify species and to explain how to collect data



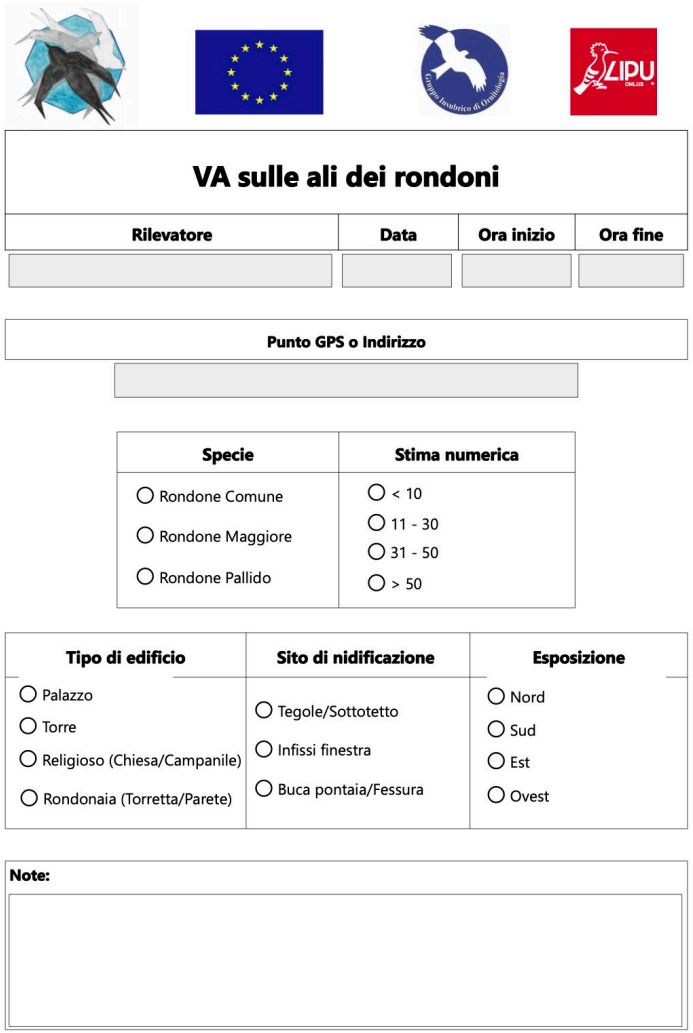
- WhatsApp group to coordinate volunteers
- Field sessions for observations and data collection

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METHODS

We collected these data for each colony:

- GPS coordinates
- Species
- Abundance class
- Kind of building
- Kind of site
- Orientation



VA sulle ali dei rondoni

Rilevatore	Data	Ora inizio	Ora fine

Punto GPS o Indirizzo

Specie	Stima numerica
○ Rondone Comune	○ < 10
○ Rondone Maggiore	○ 11 - 30
○ Rondone Pallido	○ 31 - 50
	○ > 50

Tipo di edificio	Sito di nidificazione	Esposizione
○ Palazzo	○ Tegole/Sottotetto	○ Nord
○ Torre	○ Infissi finestra	○ Sud
○ Religioso (Chiesa/Campanile)	○ Buca pontaia/Fessura	○ Est
○ Rondonaia (Torretta/Parete)		○ Ovest

Note:

METHODS



Analysis for the dataset

- QGIS 2.12 for the map
- Excels for simple diagrams



- *Apus apus*:
  - Python programming language (Google Colab)
  - One-way anova test
  - Pearson correlation index

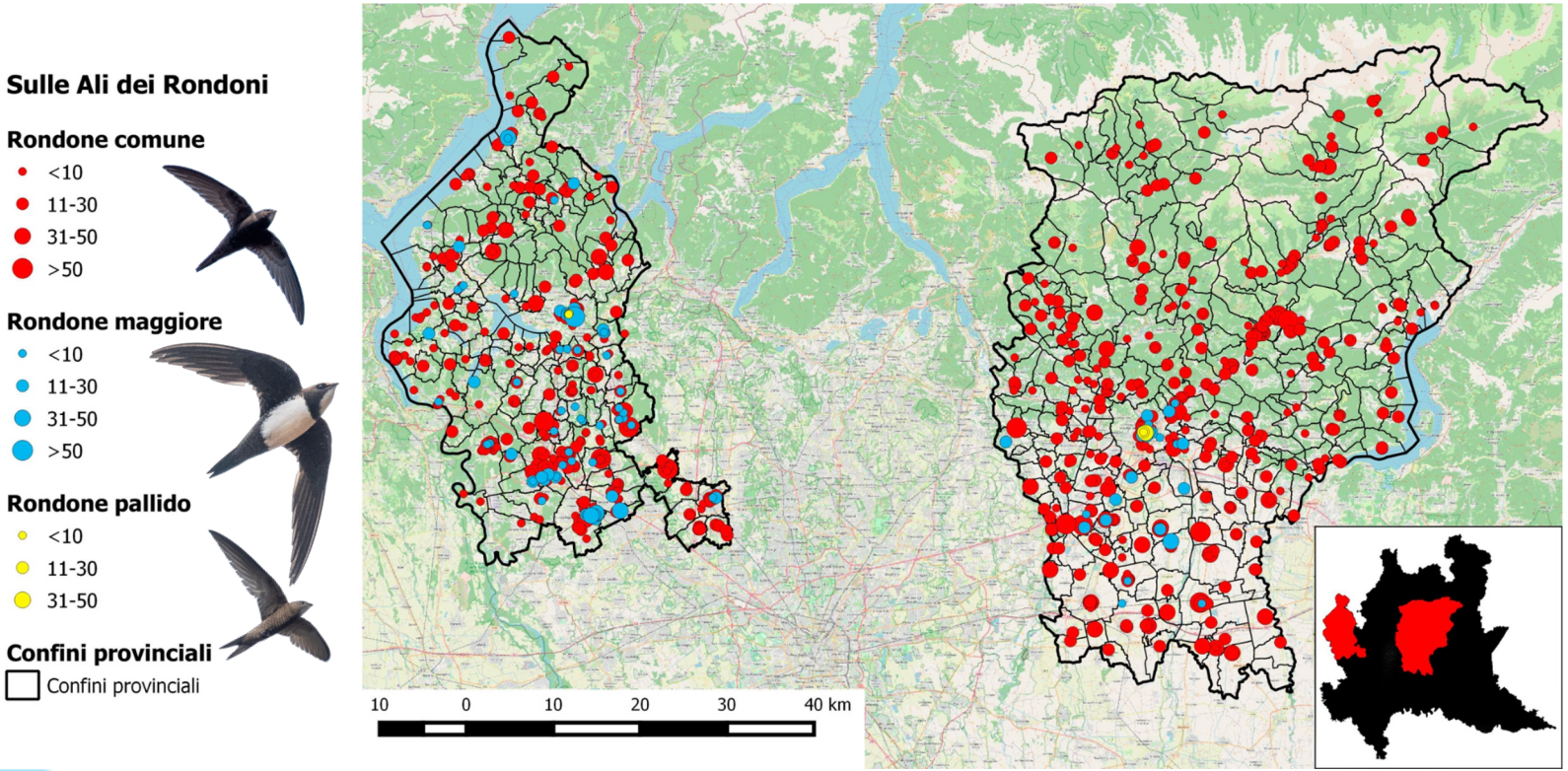


- *Tachymarptis melba*:
  - Order logit model → logit(Y ≤ j) = b0 + b1x1… + bpxp
  - Software R 4.0.5 with polr function in MASS
  - Does distribution change with the dimension of the cities? We found density data from ISTAT website.

RESULTS



RESULTS



992 colonies found:

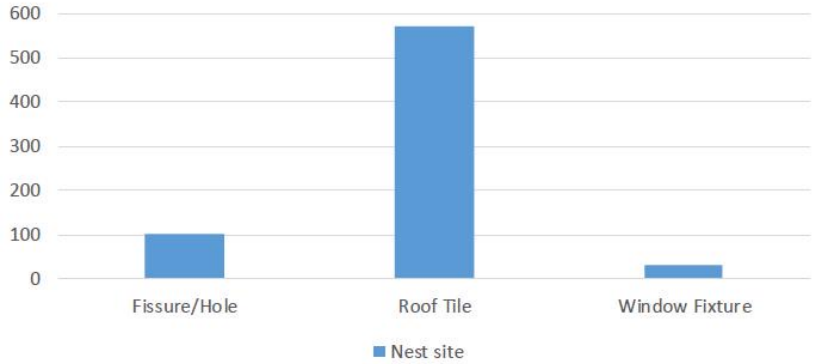
- 86,6% Common Swift - 12,8% Alpine Swift - 0,6% Pallid Swift

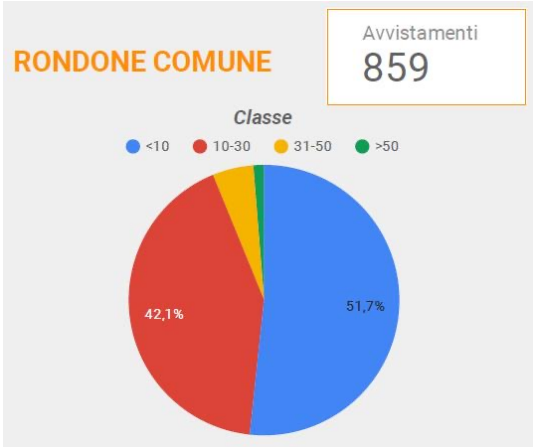
RESULTS

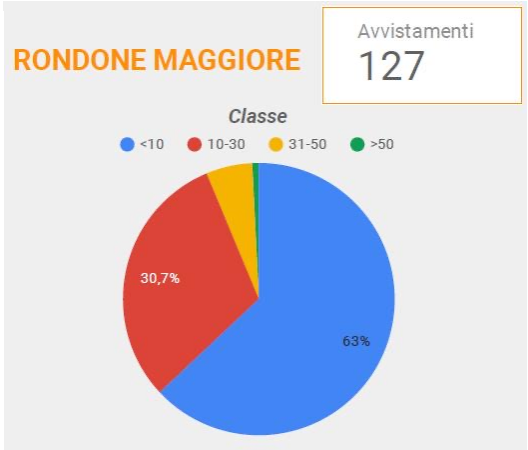
Most used sites:

roofs (81,1%)



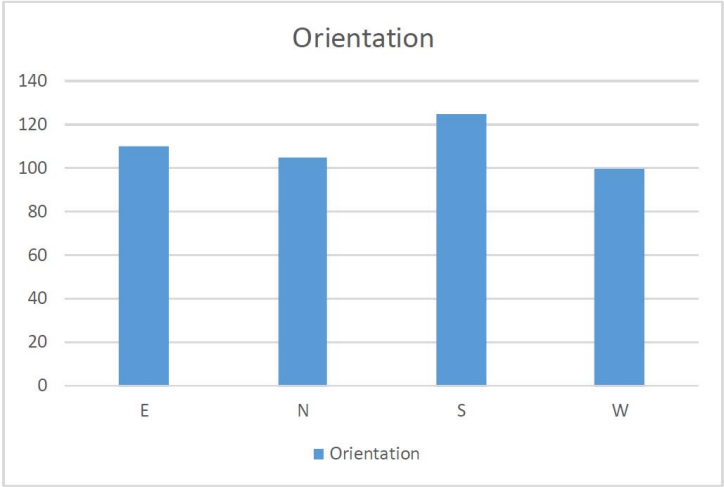






RESULTS – *Apus apus*

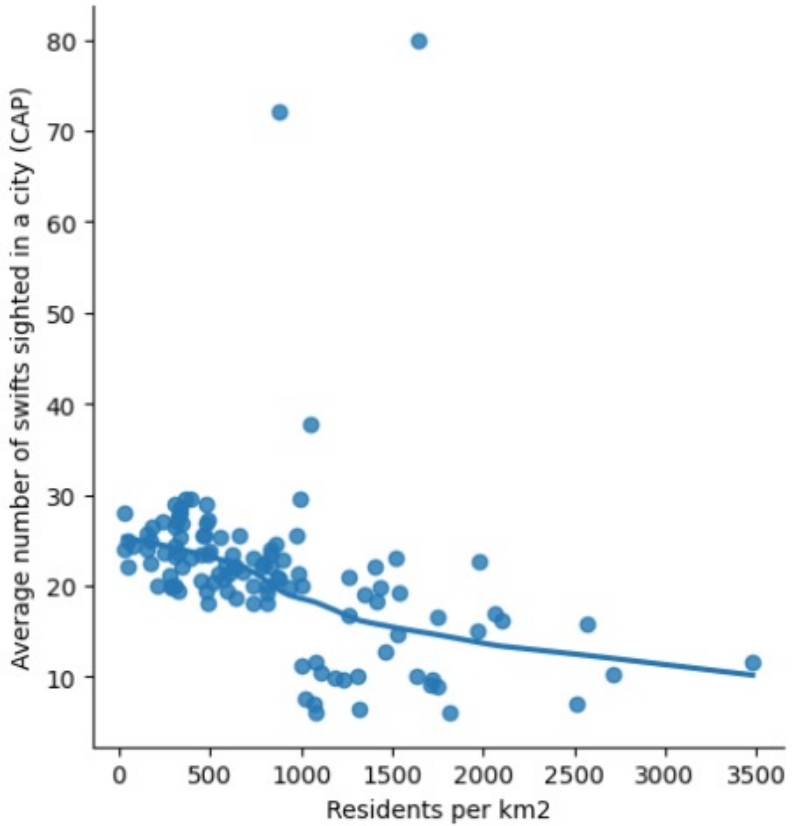
- 94% of colonies with less of 30 individuals
- The Tukey HSD post-hoc comparison showed that North faces hosted significantly less nests than East faces, while the other ones did not differ significantly to one another



Group 1	Group 2	Difference in mean values	P-adj
E	N	-3.60	0.04
E	S	-1.80	0.52
E	W	-3.38	0.07
N	S	1.81	0.52
N	W	0.23	0.91
S	W	-1.58	0.63

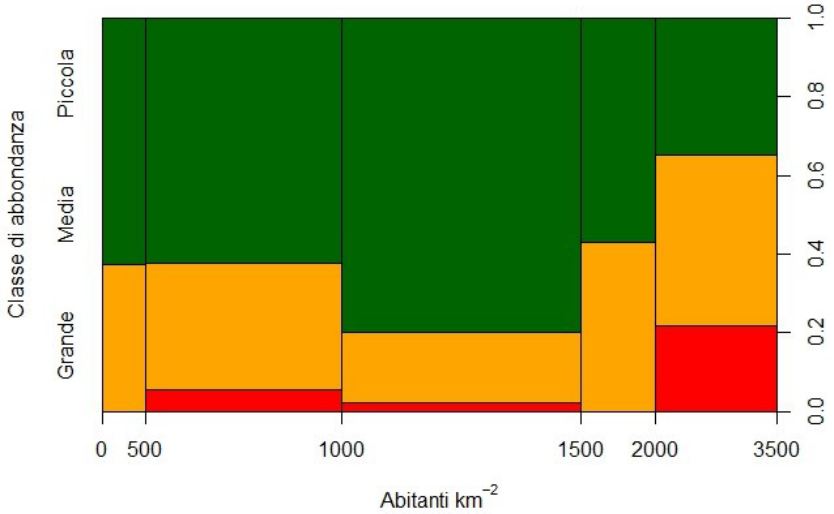
RESULTS – *Apus apus*

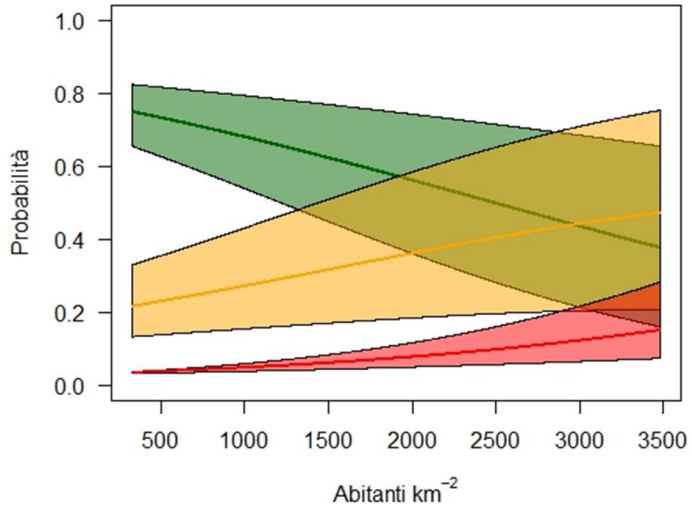
- Pearson correlation index: coefficient = -0.3
- No strong linear relationship between the population density of the cities and the number of Swifts
- We represent the descriptive trend of their relationship using a polynomial



RESULTS – *Tachymarptis melba*

	value	standard error	t value	p value
I(abitanti/1000)	0.507	0.229	2.218	0.027
piccola | media	1.261	0.385	3.275	0.001
media | grande	3.490	0.527	6.625	< 0.001





Small colonies (green) are more abundant in small to medium-sized urban centers (Likelihood ratio test: $\chi 2 = 5.00$, degrees of freedom = 1, $p = 0.0254$)

The probability that there are small colonies in a town decreases as the population density of the latter increases.

DISCUSSION

- Importance of Swift towers (according to Ferri, 2018)
- Roof tiles are the most used site (according to Boano, 2012 and Pirovano et al., 2022)
- No orientation preferences (according to Galeotti, 1993 and Bronne, 2009)
- Interesting preference for the less densely populated cities
- Data could be used to expand the knowledge about local Swift ecology
- Common Swift seems to nest mainly in civil buildings (Milani, 2022), while no significant difference was observed for the Alpine Swift in the type of occupied buildings, nor for the breeding site (Iadanza, 2023)

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DISCUSSION

- Some curious sites: cranes
- Raise public awareness
- Shapefiles to municipalities to protect colonies



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- BiodiverCities Program of European Commission
- All the authors of the photos

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Thanks for your attention

